

BUSINESS REQUIREMENTS

The Motor Insurance System should support the following business requirements:

1. Users should log in using an encrypted password.
2. Customers can create a new insurance quote.
3. Customers can choose either:
 - Comprehensive Coverage
 - Third Party Liability (TPL)
4. Premium should be calculated based on the selected coverage plan.
5. Available add-on coverages and benefits should be displayed.
6. Customers can select one or more add-on coverages.
7. Final premium should include taxes (GST).
8. The system should support payments in foreign currencies.
9. If the customer is a broker with a prepaid credit account, the premium amount should be deducted from the broker's balance.
10. After successful payment, a unique policy number should be generated.
11. Policy numbers should follow the insurance company's standard format.
12. Credit Note and Debit Note should be generated automatically.
13. Policy Schedule should be generated after policy issuance.

DESIGN

MASTER TABLES

Table Name	Purpose
vehicle_make	Stores vehicle manufacturer details
vehicle_model	Stores vehicle model details
states	Stores state information
cities	Stores city information
vehicle_color	Stores available vehicle colors
vehicle_body	Stores vehicle body types

Table Name	Purpose
vehicle_category	Stores vehicle categories
product_config	Stores insurance products and coverage plans
addon_cover	Stores additional insurance coverages
premium_rate	Stores premium calculation rates
lov	Stores common values like Gender, Education, Currency, Payment Mode

TRANSACTION TABLES

Table Name	Purpose
users	Stores customer personal information
login_users	Stores login credentials
brokers	Stores broker information
customer_vehicle	Stores customer vehicle details
quotes	Stores quotation details
quote_addons	Stores selected add-on coverages
payments	Stores payment transactions
policies	Stores issued insurance policies
broker_transactions	Stores broker credit/debit transactions
commission	Stores broker commission details
credit_note	Stores generated credit notes
debit_note	Stores generated debit notes
policy_schedule	Stores generated policy schedule details

SEQUENCE REQUIRED

Since MySQL doesn't support Oracle-style sequences directly, these can be implemented using AUTO_INCREMENT.

Sequence	Description
Customer ID	Generates unique customer IDs
User ID	Generates unique login user IDs
Quote Number	Generates unique quotation numbers
Policy Number	Generates unique policy numbers
Payment ID	Generates unique payment transaction IDs
Broker ID	Generates unique broker IDs
Vehicle ID	Generates unique vehicle IDs

SAMPLE TABLE STRUCTURES

VEHICLE_MAKE

Column	Data Type	Constraint
make_id	INT	PRIMARY KEY, AUTO_INCREMENT
make_desc	VARCHAR(100)	NOT NULL
status	VARCHAR(20)	NOT NULL
added_on	DATE	DEFAULT CURRENT_DATE
added_by	VARCHAR(50)	NOT NULL

VEHICLE_MODEL

Column	Data Type	Constraint
model_id	INT	PRIMARY KEY, AUTO_INCREMENT
model_desc	VARCHAR(100)	NOT NULL
make_id	INT	FOREIGN KEY

Column	Data Type	Constraint
status	VARCHAR(20)	NOT NULL
added_on	DATE	DEFAULT CURRENT_DATE
added_by	VARCHAR(50)	NOT NULL

USERS

Column	Data Type
user_id	INT
first_name	VARCHAR(50)
last_name	VARCHAR(50)
gender	VARCHAR(20)
dob	DATE
email	VARCHAR(100)
phone	VARCHAR(15)
mobile	VARCHAR(15)
address1	VARCHAR(100)
address2	VARCHAR(100)
street	VARCHAR(100)
city_id	INT
state_id	INT
national_id	VARCHAR(20)
nationality	VARCHAR(50)
status	VARCHAR(20)
added_on	DATE

Column	Data Type
added_by	VARCHAR(50)

LOGIN_USERS

Column	Data Type
login_id	INT
username	VARCHAR(50)
password	VARCHAR(255)
user_id	INT
lead_id	INT
user_role	VARCHAR(30)
status	VARCHAR(20)
added_on	DATE
added_by	VARCHAR(50)

BROKERS

Column	Data Type
broker_id	INT
broker_name	VARCHAR(100)

Column	Data Type
organization_name	VARCHAR(100)
contact_no	VARCHAR(15)
email	VARCHAR(100)
credit_balance	DECIMAL(10,2)
commission_percentage	DECIMAL(5,2)
status	VARCHAR(20)
added_on	DATE
added_by	VARCHAR(50)

IMPLEMENTATION FLOW

User Login



Create Customer



Register Vehicle



Select Insurance Plan



Choose Add-on Covers



Calculate Premium



Apply GST



Make Payment



Generate Policy



Generate Credit/Debit Note



Generate Policy Schedule